Urban GeoBIM Construction by Integrating Semantic LiDAR Point Clouds With As-Designed BIM Models

Jie Shao, Wei Yao, *Senior Member, IEEE*, Puzuo Wang, Zhiyi He, and Lei Luo

***Abstract*— Developments in 3-D real worlds promote the integration of geoinformation and building information models (BIMs) known as GeoBIM. Light detection and ranging (LiDAR) integrated with global navigation satellite systems (GNSSs) can provide geo-referenced information at the urban scale. However, constructing detailed urban GeoBIM poses challenges in terms of LiDAR data quality. BIM models designed from software are fine on geometrical information but are often located in local coordinates and limited at an individual building level. In this study, we propose a complementary strategy to solve the contradiction between fine and large-scale construction of GeoBIM in urban scenes. A deep learning framework and graph theory are combined for LiDAR point cloud segmentation. Then, a coarse-to-fine matching program is developed to integrate building point clouds with corresponding BIM models. Results show that the overall segmentation accuracy of LiDAR datasets reaches up to 90%, and average positioning accuracies of building BIM models are 0.156 m, demonstrating the effectiveness of the method in segmentation and matching processes. This work offers a practical solution for rapid and accurate urban GeoBIM construction.**

***Index Terms*— Building information model (BIM), light detection and ranging (LiDAR) point cloud, matching, segmentation, urban GeoBIM.**

I. Introduction

URBAN scene is a dynamic environment that comprises buildings and other infrastructures, making large-scale 3-D urban construction crucial for various applications, such as urban design, transportation, and tourism [1]. With advancements in information technology, an increasing number of technologies for reconstructing urban scenes are being proposed, with digital twins and the 3-D real scenes being the current research hotspots in the field [2], [3]. These technologies describe physical entities in the real world into virtual spaces through digitization [4]. By leveraging digitized entities, urban infrastructures can be accurately understood and provide valuable references for urban management and decision-making.

Building information modeling (BIM) is a widely used digital tool for urban scene construction, integrating geometry, attributes, and status information throughout the whole life cycle of infrastructure, from engineering design to construction and operation [5]. Nowadays, there are numerous software and tools, such as Autodesk Revit, related to BIM technologies that have been developed and applied in 3-D urban construction for various purposes, such as hazard analysis [6] and environmental impact assessment [7]. However, BIM models created in software manually are often represented based on a local coordinate system, lacking accurate geo-referenced locations, which poses challenges for accurate urban construction [8]. In recent years, some engineering projects and academic studies have highlighted the advantage of using geo-referenced BIM models for urban planning and asset management [9], [10]. As a result, the integration of geoinformation and BIM models, known as GeoBIM, is gaining increasing attention within the urban construction community [11].

As mentioned earlier, BIM models for infrastructures are typically endowed with geoinformation through manual input using human–computer interaction. However, this approach can be time-consuming and less accurate compared to using 3-D real scene data. To address this issue, some studies have proposed inverse procedural modeling approaches [12], where BIM models are generated from geographic information system (GIS) data. Light detection and ranging (LiDAR) integrated with a global navigation satellite system (GNSS) is a powerful tool that can directly provide accurate geospatial GIS data, i.e., geo-referenced 3-D point cloud [13]. By leveraging modeling technology, LiDAR point clouds can be meshed to reconstruct multiple levels of detail (LoDs) urban models [14], [15]. However, existing methods for LiDAR-based urban construction often have strict requirements on data quality, including accuracy and integrity. Single-platform LiDAR systems may struggle to provide complete spatial informa-

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Jie Shao is with the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong, China, and also with the Hong Kong Polytechnic University Shenzhen Research Institute, Shenzhen 518000, China (e-mail: jie.shao@polyu.edu.hk).

Wei Yao is with the Department of Land Surveying and Geo-Informatics and Otto Poon Charitable Foundation Smart Cities Research Institute, The Hong Kong Polytechnic University, Hong Kong, China, and also with the Hong Kong Polytechnic University Shenzhen Research Institute, Shenzhen 518000, China (e-mail: wei.hn.yao@polyu.edu.hk).

Puzuo Wang and Zhiyi He are with the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong, China (e-mail: puzuo.wang@connect.polyu.hk; zhiyi.he@polyu.edu.hk).

Lei Luo is with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100045, China (e-mail: luolei@aircas.ac.cn).

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tion for complex urban scenes, and challenges may arise in acquiring data from difficult-to-reach observation locations or fusing data from multiplatform LiDAR systems, resulting in inaccurate and incomplete urban models. Additionally, most previous studies have focused on simple-shaped entities, with limited attention to complex entities [16]. Therefore, there is a need for further exploration of efficient and high-quality urban construction.

To address the research gap, this study proposes a rapid and accurate method for urban GeoBIM construction by integrating LiDAR point cloud with as-designed BIM models. Specifically, LiDAR data and as-designed BIM models are first collected for study areas. A deep learning framework is used for the semantic segmentation of LiDAR point clouds, in which an adaptive method is used for automatically labeling training samples to increase efficiency; a graph-based clustering approach combining multiple geometric attributes is then adopted to instance and optimize semantic segmentation, which has the potential to improve the segmentation accuracy. Next, a coarse-to-fine matching strategy is developed to integrate LiDAR point clouds with the corresponding as-designed BIM models, especially proposing an automated approach for coarse alignment of BIM models with sparse point clouds and LiDAR data of buildings through nonplanar four-point pairs. This study aims to offer a fresh perspective on urban GeoBIM construction by leveraging the principles of complementary advantages and maximizing the utilization of accurate geo-referenced information of large-scale LiDAR point cloud and detailed geometric attribute information of BIM models, thereby solving the contradiction between detailed and large-scale urban GeoBIM construction. Following the introduction section, Section II presents a comprehensive literature review of the related works on point cloud segmentation and matching of GIS data and BIM models. The proposed method will be described in detail in Section III. Experimental results will be introduced in Section IV and discussed in Section V, and then, conclusions are presented.

II. Literature Review

A. Three-Dimensional Point Cloud Segmentation

The basic premise of this study is to segment individual objects from urban scenes and provide geo-referenced data sources for the corresponding BIM models of each infrastructure. Conventional approaches, such as region growing [17] and DBSCAN [18], typically segment point clouds based on geometric characteristics, such as planar and cylinder shapes. As a result, these methods are generally suitable for simple objects with regular geometric shapes [16] that can be represented by fixed parameters. However, complex-shaped objects are difficult to segment based on single geometric features, as they often require cumbersome rule-based designs, are sensitive to data sources, and lack generalization capabilities. While there are some methods that can represent complex geometric shape features [19], [20], obtaining accurate individual objects solely through traditional segmentation methods in urban scenes remains difficult.

Recent advancements in machine learning and deep learning technologies have enabled more intelligent extraction of rich semantic information from point clouds. Traditional machine learning methods often require a manual definition of geometry or topology features, followed by the use of classifiers such as support vector machine and random forest to label points [21], [22]. Deep learning technologies can automatically construct classification models and classify objects by directly inputting point clouds into a deep neural network, without handcrafted geometry or topology features. It has become the dominant method in current point cloud classification [23], with the introduction of the PointNet [24] framework setting a precedent in this field. The PointNet framework addresses challenges such as point cloud disorder and permutation and rotation invariances with low cost. However, it ignores the relationship between neighboring points in the local field. To overcome this limitation, an expanded framework called PointNet++ [25] was proposed, which incorporates local hierarchical modules to improve segmentation accuracy by considering features in local and global receptive fields. Inspired by these deep learning frameworks, several other networks with different strategies have been proposed for 3-D point cloud segmentation, including convolutional-based networks such as RS-CNN [26] and KPConv [27], graph-based networks such as PointGCN [28], and deep feature embedding [29].

The aforementioned networks commonly perform well in semantic segmentation of point clouds but still struggle with segmenting individual instances, which has led to the recent study of higher level instance segmentation technologies [16], such as SGPN [30], PointGroup [31], and OccuSeg [32]. These methods provide better references for intelligent segmentation of individual object point clouds. However, these deep learning frameworks are often reliant on manually labeled training samples, which can be a time-consuming process in large-scale scenes. Moreover, most current instance segmentation methods rely on point cloud clustering based on semantic segmentation results, such as graph-based instance clustering strategies [31], [32]. These works typically consider the distance between neighboring points or voxels to segment instances from the same semantic class, resulting in clustering results that are sensitive to the semantic segmentation and distance threshold settings. Incorrect semantic class assignments or improper threshold settings can adversely affect clustering qualities and the final integration of LiDAR data with BIM models.

B. Integration of GIS Data and BIM Model

The integration of GIS and BIM aims to bring as-designed BIM models, which are typically in local coordinate systems, into a GIS environment [9]. Conventional approaches involve the manual definition of geo-referenced information for BIM models during the design phase using software [33]. However, the accuracy of GIS and BIM integration can be subjective and prone to errors introduced during manual operations. Additionally, existing methods struggle to generate standardized solutions that are scalable for integrating many BIM models into the geo-referenced environment.

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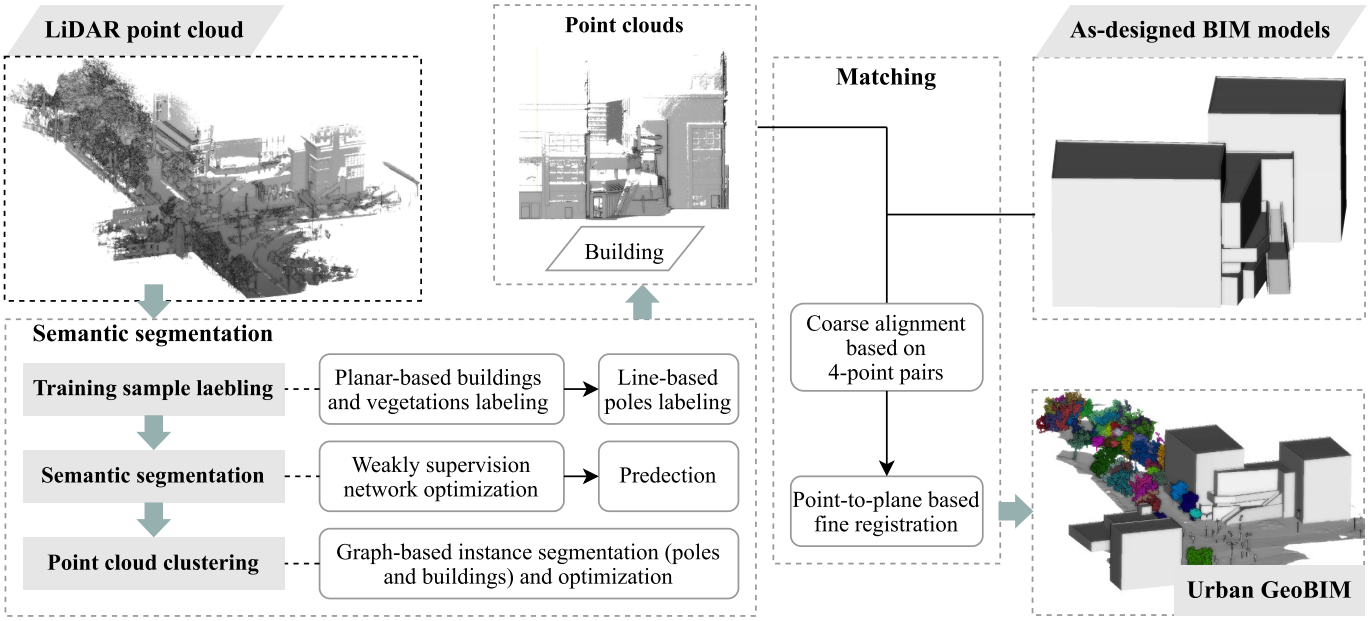


Fig. 1. Workflow of this article, including preparation of LiDAR point clouds and as-designed BIM models, semantic segmentation of LiDAR point clouds, and coarse alignment and fine registration of instance point clouds and BIM models.

In practice, the integration of GIS data, such as LiDAR point clouds, and as-designed BIM models can be addressed through data matching techniques commonly used in computer graphics and remote sensing communities [34], [35]. The core of matching is to establish uniform coordinate systems for different data by accurate correspondences [36], [37]. Typically, the corresponding relationship between two spatial data can be represented by rigid-body transformations involving rotation, translation, and scaling, where geometric features are often detected to calculate transformation parameters. For instance, some researchers have projected 3-D building data into a 2-D plane along the z-axis to generate planar features of building footprints for matching 3-D GIS data and the corresponding BIM models [10]. These methods perform well for buildings with irregular or unique footprints but face challenges for buildings with similar footprints, in which case 3-D features may provide more information for matching. However, there are limited automated approaches and common methods still rely on manual operations [33]. BIM models are typically described by polygon meshes, where each mesh consists of multiple connection points. Hence, the matching of LiDAR data and BIM models can also be considered as point cloud registration.

Currently, there are automated registration algorithms based on feature descriptors that have been proposed for point clouds, such as point feature histograms (PFHs) [38] and scale-invariant feature transform (SIFT) [39] algorithms, which are typically suitable for continuously distributed point clouds. However, BIM models are composed of discrete isolated points and may not provide effective feature descriptors for registering with LiDAR point clouds. In contrast, the state-of-the-art algorithm, four-point congruent sets (4PCSs), matches almost-planar four-point basis in two inputs [40], [41]. This algorithm detects features from a global scale without considering the distribution of local points and can provide a reference for the registration of LiDAR data and BIM models. These algorithms are typically applied to coarse alignment and may not obtain accurate registration results. The conventional algorithm for fine registration is the iterative closest point (ICP) algorithm, which registers point clouds by minimizing the distance between pairs or corresponding points in two datasets [42]. However, this method may not be suitable for datasets with low overlap, as it relies on rich structural information. Therefore, further research is needed to develop fine registration for LiDAR data and BIM models.

III. Methodology

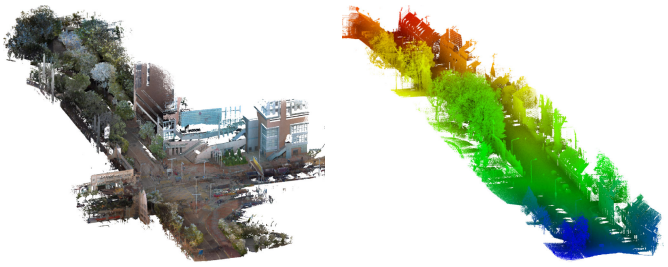
A. Overview

This study proposes an urban GeoBIM construction strategy that integrates geo-referenced LiDAR point clouds and as-designed BIM models, which involves semantic segmentation of urban objects and matching of LiDAR data and BIM models (Fig. 1). Given the need for the management of fixed infrastructure equipment, the focus of this work is GeoBIM construction of urban buildings. In the semantic segmentation process, an automated labeling strategy of training samples is first adopted to detect objects from LiDAR point clouds. Then, a weakly supervised framework is used for semantic segmentation; individual objects, meanwhile, are segmented based on a "splitting and merging" strategy. Last, a coarse-to-fine registration is employed to match LiDAR point clouds and as-designed BIM models of buildings.

B. Data Preparation

Two datasets, namely Dataset 1 and Dataset 2, from different urban locations, are selected for evaluating the effectiveness of this method. Dataset 1 was acquired from Hong Kong, China, while Dataset 2 was obtained from Lille, France. In each dataset, geo-referenced LiDAR data were collected

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(a) (b)

Fig. 2. Two test datasets. Colored LiDAR point cloud in (a) Dataset 1 and (b) Dataset 2.

using mobile laser scanning systems, and the corresponding BIM models were created manually.

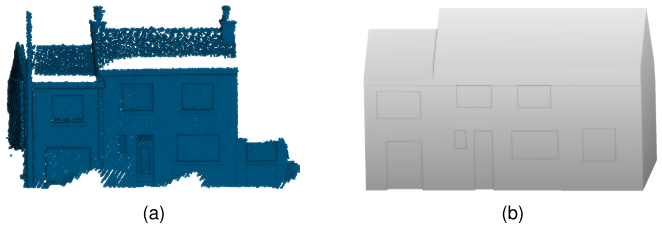
1) LiDAR Point Clouds: In Dataset 1, the LiDAR point cloud was acquired using a backpack LiDAR system that integrates an inertial measurement unit (IMU), GNSS, and a laser scanner (VELODYNE VLP-16). The system provides centimeter-level position accuracy, with a maximum scan range of up to 100 m, a field of view of 360° × 30°, and angle resolutions of 0.2° in the horizontal direction and 2° in the vertical direction. The scene area of Dataset 1 is 235 × 225 m, and the point cloud resolution is approximately 0.01 m. However, due to the horizontal placement of the laser scanner, acquiring superstructure information from the backpack LiDAR system in Dataset 1 is challenging [Fig. 2(a)]. To evaluate the classification result, we manually annotate the data with five categories, including ground, building, vegetation, pole-like object, and others. Dataset 2 is obtained from the open urban point cloud dataset Paris-Lille-3-D [43], and the point cloud has a high density between 1000 and 2000 points per square meter on the ground. For this study, a street segment approximately 360 m in Lille, France, was selected for validating the proposed method [Fig. 2(b)]. Additionally, to increase efficiency, LiDAR data are subsampled with a grid size of 0.03 m. Except for maintaining points of ground, buildings, and natural (vegetation) and pole-like objects, we assigned the remaining points as the category of others.

2) BIM Models: BIM models of buildings were designed using Autodesk 3DMAX, a commercial software. However, due to the lack of as-designed BIM models, we referred to the LiDAR point cloud [Fig. 3(a)] to design coarse BIM models [Fig. 3(b)] for verification purposes. Given the characteristics of LiDAR data, buildings were designed as LoD2 models.

C. Semantic Segmentation of LiDAR Point Cloud

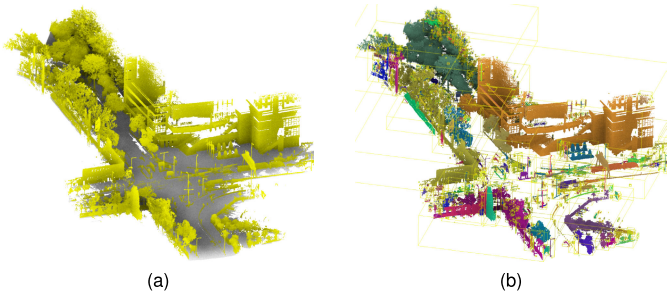
To detect individual semantic objects from urban point clouds, we implement three segmentation processes that combine training sample labeling, semantic segmentation, and point cloud clustering, in which urban objects include unclassified (others), buildings, vegetation, ground, and pole-like (such as traffic sign posts and streetlight) categories.

1) Training Sample Labeling: As a supervised classification strategy is adopted in this study, the labeling of training samples is a primary challenge to be addressed. Manually labeling training samples in urban scenes can be laborious and time-consuming due to the complexity of the scenes. In practice, after filtering out the ground points, some objects in urban scenes may be isolated and not connected to others. Additionally, different objects typically exhibit unique geometric properties: for example, pole-like objects can be described by lines, buildings can be described by planes, and vegetation points show discrete characteristics. Thus, an adaptive automated labeling method based on geometric attributes is presented in this article. Specifically, ground points are detected using the cloth simulation filter algorithm [44], and then, the nonground points are clustered using the connected component labeling algorithm (Fig. 4). We select the points in the lower half of each cluster to conduct spatial line fitting [Fig. 5(a)]. If 90% of the points that participate in line fitting have a minimum spatial distance of 0.5 m from the fit line, the cluster will be labeled as a pole-like object.



(a) (b)

Fig. 3. As-designed BIM models. (a) Raw point cloud of a building. (b) Corresponding BIM model.



(a) (b)

Fig. 4. (a) Ground filtering and (b) nonground points clustering of Dataset 1. In (a), gray points represent the ground, yellow points represent nonground, and random colors in (b) represent different clusters.

The local surface of the building is typically smooth, while that of vegetation is rough. Therefore, surface roughness is used as a criterion for determining buildings and vegetation. Furthermore, due to the connectivity of these clusters, they often have large sizes and point amounts. For a point $p_i(x_i, y_i, z_i)$ in these clusters, the point roughness r_i can be calculated based on the eigenvalues of a covariance matrix $C_{3\times3}$, where C is defined as

$$C = \frac{1}{n}\sum_{i=1}^{n}\left(p_i - \bar{p}\right)\left(p_i - \bar{p}\right)^T \tag{1}$$

where $\bar{p} = ((1/n))\sum_{i=1}^{n} p_i$ is the average of n nearest neighboring points of p_i, and then r_i is defined as

$$r_i = \frac{\lambda_2}{\lambda_0 + \lambda_1 + \lambda_2} \tag{2}$$

where $(\lambda_0, \lambda_1, \lambda_2)$ are eigenvalues of C sorted in descending order as $\lambda_0 \geq \lambda_1 \geq \lambda_2$. To ensure accurate labeling, strict thresholds are set to determine semantic information; if the

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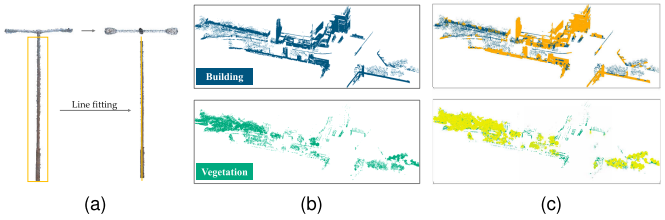


Fig. 5. Training sample labeling. (a) Pole-like object labeling. (b) Building and vegetation labeling before voxelization. (c) Building (orange points) and vegetation (yellow points) labeling after voxelization.

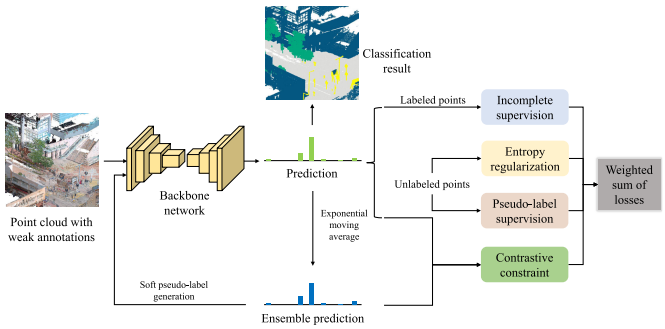


Fig. 6. Pipeline of the weakly supervised method, including entropy regularization, contrastive constraint, and online soft pseudolabeling.

roughness r_i of a point is less than 0.1, it will be considered a building point, and if the roughness r_i is greater than 0.15, the point will be considered as a vegetation point. However, the strategy may result in a loss of detailed information and underfitting of deep learning networks due to a lack of data [Fig. 5(b)]. To address this, this article further labels objects by voxelization. The raw point cloud is voxelized with a size of 2 m, and when there is only one labeled class in a voxel and its adjacent voxels have no labeled points or only points of the same class, then all points in the voxel are labeled as that class [Fig. 5(c)]. Moreover, other objects in the scene, such as vehicles and humans, maybe not part of the infrastructure and are structurally complex. In such cases, these objects are manually labeled as a separate class, i.e., others.

2) Weakly Supervised Semantic Segmentation: Given that reliable labels are available for only a fraction of points, we incorporate a weakly supervised framework with a deep learning-based neural network to segment point clouds as described in our previous work [45], which is illustrated in Fig. 6. In this framework, KPConv [27] is utilized as the backbone network. As in fully supervised learning, we first apply cross-entropy loss calculation and backpropagation to weakly labeled points, formulated as

$$\mathcal{L}_{\text{seg}} = -\frac{1}{M}\sum_{i}^{M}\sum_{c}^{K} y_{ic}\log z_{ic} \tag{3}$$

where z_{ic} and y_{ic} are the predicted probability and corresponding label, respectively. To employ latent information from unlabeled data, three weakly supervised constraints are adopted to learn accurate point cloud semantics.

Specifically, entropy regularization [46] is first incorporated to minimize the uncertainty of the posterior probability by reducing the Shannon entropy H of pointwise predictions, presented as

$$\mathcal{L}_{\text{ent}} = -\frac{1}{N-M}\sum_{i}^{N-M} H_i$$
$$H_i = \sum_{c}^{K} z_{ic}\log z_{ic}. \tag{4}$$

This is based on the finding that unlabeled data contain less information with high uncertainty when prediction classes overlap, while the Shannon entropy is prevalent in quantifying the amount of information contained in the data. Furthermore, inspired by contrastive learning, we introduce consistency constraints to produce stable predictions by enforcing consistency between the current semantic prediction z and its ensemble value $\tilde{z}(t) = \alpha \cdot \tilde{z}(t-1) + (1-\alpha)\cdot z$. We use mean square error (MSE) for consistency loss calculation

$$\mathcal{L}_{\text{epc}} = -\frac{1}{N}\sum_{i}^{N} |z_i - \tilde{z}_i|^2. \tag{5}$$

Online soft pseudolabeling is used to expand the labeled pool in an efficient and nonparametric way. Pseudolabels are defined as the category with the maximum prediction probability of ensemble predictions $y_i^{pl} = \arg\max \tilde{z}_i$. Instead of empirically setting a fixed threshold for pseudolabel selection, we leverage all unlabeled points and soften their pseudolabels with reliability weights w that are directly inferred from prediction entropy

$$\mathcal{L}_{\text{seg}}^{pl} = -\frac{1}{N-M}\sum_{i}^{N-M} w_i \sum_{c}^{K} y_{ic}^{pl}\log z_{ic}$$
$$w_i = 1 - H_i/\log K. \tag{6}$$

All the proposed losses jointly participate in backpropagation during training, enhancing model performance without sacrificing efficiency.

3) Graph-Based Point Cloud Clustering: Different objects can be distinguished based on the semantic segmentation process, but obtaining individual objects still requires the instance segmentation process. In urban scenes, buildings and pole-like objects are often isolated, but individual object points are continuously distributed. Therefore, a "splitting and merging" strategy is used to segment individual objects and optimize semantic segmentation results. The conventional graph-based clustering approach typically segments the point cloud based on distance information, which performs well for isolated objects but struggles with connected objects. To enhance applicability, a combination of topological and geometric relationships between points is proposed for graph-based clustering, where the geometric relationship is described by roughness and anisotropy. The connected component labeling algorithm is first used to split each semantic class into discrete point clusters. Subsequently, an undirected graph is constructed to merge these point clusters, with each cluster represented as a vertex and connections between vertices referred to as edges. The minimum distance, average roughness, and anisotropy of each cluster are calculated. The distance is calculated as the Euclidean distance, roughness is

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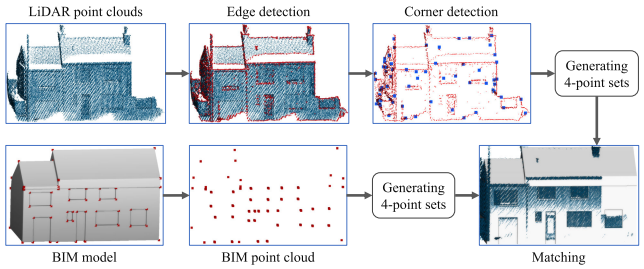


Fig. 7. Matching process of BIM model and LiDAR point cloud of a building, involving edge detection and corner detection of LiDAR point cloud, and generating and matching four-point sets.

defined as (2), and anisotropy is defined as $a_i = (\lambda_0 - \lambda_2)/\lambda_0$. The connection weight between each pair of vertices can be calculated by

$$w_{ij} = e^{-\frac{1}{3}\left(\left(\frac{d_{ij}}{\sigma_1}\right)^2+\left(\frac{r_i-r_j}{\sigma_2}\right)^2+\left(\frac{a_i-a_j}{\sigma_3}\right)^2\right)} \tag{7}$$

where w_{ij} and d_{ij} represent the connection weight and the minimum distance between vertexes i and j, respectively, and $(\sigma_1, \sigma_2, \sigma_3)$ represent the standard deviations of distance, roughness, and anisotropy. If w_{ij} is greater than a threshold (0.75), the adjacent vertices will be considered as belonging to the same cluster. In practice, the class of the cluster is defined by the class of the vertex with the largest size.

D. Matching of LiDAR Point Clouds and BIM Models

The matching of LiDAR and BIM data is the basis for constructing geo-referenced BIM in this study. Therefore, a coarse-to-fine matching process is adopted to match LiDAR and BIM point clouds of buildings (see Fig. 7). Given the sparse point clouds of BIM models, a robust coarse alignment method is first proposed, inspired by the 4PCS algorithm. The conventional 4PCS algorithm based on near-coplanar four-point sets detected from raw point clouds is sensitive to the initial position of point clouds and easily falls into the local optimal solutions, which limits the robustness. Here, we replace near-coplanar four-point sets with noncoplanar four-point sets and search for the correspondences of all four-point sets in the base point cloud, rather than just determining the optimal corresponding four-point pair based on one four-point set in the base point cloud. To increase time efficiency, building edges are first detected using the angle criterion approach [47] based on the open-source point cloud library (PCL). Keypoints are then determined by calculating the angles between a point and its neighboring points, in which the point with an angle of 60°–120° will be regarded as a candidate keypoint. Moreover, we traversed each point in candidate keypoints and regarded the candidate keypoint with an angle closest to 90° within a radius of the point as a final keypoint. The matching of LiDAR keypoints and BIM models is performed, with the LiDAR dataset as the reference and BIM models as the aligned data. Specifically, corresponding four-point pairs that are not co-planar are detected from the LiDAR and BIM point clouds. The distances between all points of the BIM model and their closest neighbors in the LiDAR data are counted, and four-point pairs that obtain the smallest distance are presented as the optimal correspondence. The pseudocode of the alignment process is shown in Algorithm 1.

Algorithm 1 Coarse Alignment of LiDAR Point Cloud and BIM Model

Input:
∗ *Key points in LiDAR data* $\boldsymbol{L}$ *and as-designed BIM point cloud* $\boldsymbol{B}$
∗ *Angle range* $\boldsymbol{\theta}$ *and distance threshold* η
Output: *Rotation and translation matrices* $[\boldsymbol{R} \in \mathbb{R}^{3\times3}, T]$

```
/* Producing 2-point sets LL and BB by distances between points */
for i ← 0 to length(L)(B) − 1 do
    for j ← i + 1 to length(L)(B) do
        d_ij = |L_i − L_j|(|B_i − B_j|)
        if d_ij > η then
            LL ← (L_i, L_j), BB ← (B_i, B_j)
/* Generating four-point sets in LiDAR LS and BIM BS */
for i ← 0 to length(LL)(BB) − 1 do
    for j ← i + 1 to length(LL)(BB) do
        Distance d and angle α between 2-point pairs
        Coplanarity of 2-point pairs
        if d > η/2 && α ∈ θ && noncoplanar then
            LS ← (LL_i, LL_j), BS ← (BB_i, BB_j)
/* Determining corresponding four-point pairs PP in LiDAR and BIM */
for i ← 0 to length(BS) do
    for j ← 0 to length(LS) do
        α_ij = |α_Bi − α_Lj|, d_ij = |d_Bi − d_Lj|
        if α_ij < 1 && d_ij < 0.5 then
            PP ← (BS_i, LS_j)
/* Calculating [R, T] of LiDAR point cloud and BIM models */
for i ← 0 to length(PP) do
    LS_i(∈ PP) = R · BS_i(∈ PP) + T
    d_LB = |LS − T(BS)|
    /* The optimal coarse alignment of LiDAR and BIM data */
    if d_LB is min then
        return [R, T]
// Notes: The length (L)(B) represents the individual lengths of L and B when traversing them separately, while the length (LL)(BB) represents the individual lengths of LL and BB when traversing them separately.
```

Accurate matching results are generally difficult to obtain by the coarse alignment process, and a fine registration process becomes necessary. Conventional fine registration algorithms like ICP may struggle to achieve accurate registration due to sparse BIM point clouds and incomplete LiDAR data [48]. Therefore, in this study, a further fine registration process based on the point-to-plane association strategy is proposed, building upon the ICP registration results. For a point P in BIM models, three points $\{Q_1, Q_2, Q_3\}$ from the raw LiDAR data that are closest to the point P and not on the same line are searched for, and the point-to-plane distance d between P and the plane formed by $\{Q_1, Q_2, Q_3\}$ can be computed by

$$d = \frac{|(P - Q_1) \times ((P - Q_2) \times (P - Q_3))|}{|(P - Q_2) \times (P - Q_3)|} \tag{8}$$

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where the matching is accurate and d will be small. Additionally, the transformation relationship between LiDAR point clouds $\boldsymbol{X}_{\text{Lidar}}$ and BIM models $\boldsymbol{X}_{\text{BIM}}$ can be defined as

$$\boldsymbol{X}_{\text{Lidar}} = \boldsymbol{R} \cdot \boldsymbol{X}_{\text{BIM}} + T \tag{9}$$

where $\boldsymbol{R}$ and T are unknowns and represent rotation and translation, which can be solved by minimizing error e

$$e = \arg\min_{e} \frac{1}{2} \sum \|d(\boldsymbol{R}, T) - 0\|^2. \tag{10}$$

Then, the Levenberg–Marquardt (L-M) algorithm is used to calculate these unknown parameters. Once $[\boldsymbol{R}, T]$ is calculated, BIM models of buildings can be given the geo-referenced information.

E. Performance Evaluation Criteria

The accuracy evaluation criteria for semantic segmentation results employed in this work involved the overall accuracy (OA) and F1 scores. OA indicates the percentage of predictions correctly classified, and the F1 score stands for the harmonic mean of the precision (p) and recall (r), formulated as

$$\begin{aligned} p &= \frac{\text{TP}}{\text{TP} + \text{FP}} \\ r &= \frac{\text{TP}}{\text{TP} + \text{FN}} \\ F1 &= 2 \times \frac{p \times r}{p + r} \end{aligned} \tag{11}$$

where TP, FP, and FN are true positives, false positives, and false negatives, respectively. For the evaluation of urban GeoBIM construction and matching of LiDAR data and BIM models, the distance between corresponding features is counted, including the minimum, maximum, average, and root MSE (RMSE)

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n} d_i^2}{n}} \tag{12}$$

where n is the number of objects and d_i is the distance between correspondences.

IV. Results

A. Semantic Segmentation Results

We present the classification map of semantic segmentation results in Fig. 8. It can be observed that the majority of points are correctly classified, indicating the effectiveness of our classification method. Detailed classification results are further illustrated in Fig. 9, where semantic boundaries are well preserved, providing effective data for instance segmentation and BIM generation. However, there are some typical errors observed in the figures. For instance, points belonging to fences, which were unclassified in this study, could be misclassified as buildings, as points associated with these two categories exhibit similar verticality. Additionally, while separated pole-like objects are well classified, those that are close to trees may be directly predicted as vegetation due to their proximity.

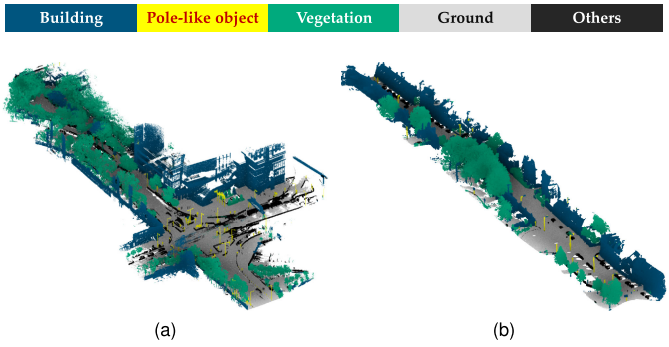


Fig. 8. Semantic classification results of (a) Dataset 1 and (b) Dataset 2.

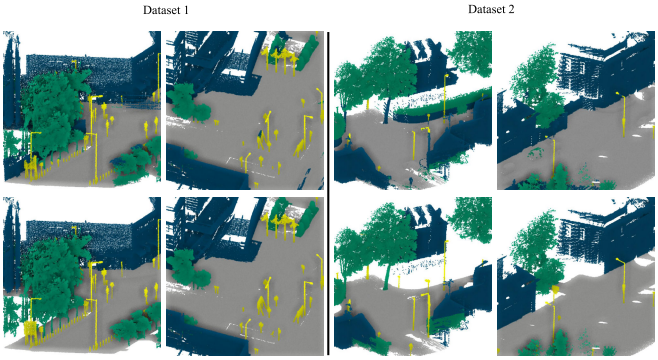


Fig. 9. Detailed classification results. The first row shows the predictions, while the second row presents the ground truth.

Furthermore, we have conducted a quantitative evaluation of classification results as shown in Table I. Categories of ground and vegetation are well classified, with high precision and recall scores. However, there are some misclassifications of unnecessary points as buildings, resulting in a relatively low precision score for the building category. This could be attributed to the limitations of MLS point cloud acquisition, where LiDAR sensors may have difficulty detecting roof areas, resulting in only the façade of buildings being preserved. As a result, some objects that were not considered in this study, such as fences, may be prone to be classified as buildings. On the other hand, due to the isolated characteristic of pole-like objects, the pole-like category shows a high precision score, although some of them may be predicted as other classes.

B. Instance Segmentation Results

To address the challenge of weakly supervision classification, where point clouds of individual objects are not available, we propose a segment-based clustering method that combines distribution characteristics of object point clouds at local and global scales. In practice, each semantic class is segmented in turn, and segments of each class are clustered based on the proposed method. Subsequently, the primary clusters of all classes are clustered again, as illustrated in Fig. 10. This approach allows us to obtain reliable clusters of objects and serves as a key step toward instance segmentation and urban GeoBIM construction.

From the results, buildings, pole-like objects, and some isolated trees can be segmented effectively. Buildings and

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TABLE I
CONFUSION MATRIX OF CLASSIFICATION RESULTS WITH PER-CLASS PRECISION, RECALL, AND F1 SCORE (%)

Dataset	Criteria (%)	Classes					Avg. F1 (%)	OA (%)
		Others	Ground	Building	Vegetation	Pole-like		
1	Precision	99.08	97.25	71.15	96.68	95.30	85.00	90.48
	Recall	39.81	99.77	99.70	96.28	85.62		
	F1 score	56.80	98.49	83.04	96.48	90.20		
2	Precision	87.61	99.08	80.83	89.20	98.04	77.55	91.24
	Recall	37.49	99.29	98.39	90.71	40.53		
	F1 score	52.51	99.18	88.75	89.95	57.35		

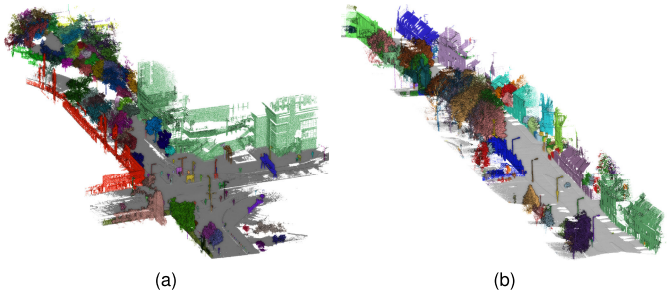


Fig. 10. Instance segmentation results of (a) Dataset 1 and (b) Dataset 2; different colors represent different instances.

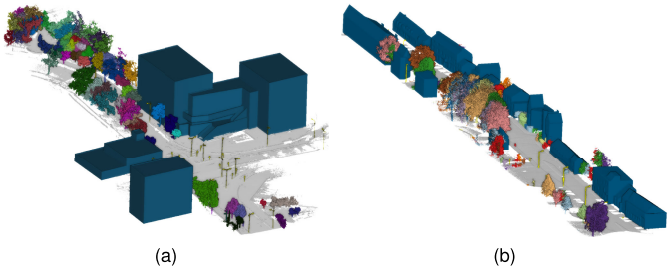


Fig. 11. Urban GeoBIM construction of (a) Dataset 1 and (b) Dataset 2.

pole-like objects, being discretely distributed in most cases, can be accurately instantiated based on the spatial distance relationship. Additionally, roughness features can also aid in correctly classifying buildings and trees that are in contact with each other. However, for gathered trees, the proposed graph-based method may not be effective for segmentation. Since the vegetation category is not considered in this study, the analysis of vegetation instance segmentation is deferred for now. When it comes to pole-like objects, errors mainly arise from the semantic segmentation results. If a pole-like object is misclassified as vegetation due to its proximity to a tree, it may be difficult to optimize and instance the pole-like object based on the proposed method. Similarly, segmenting buildings that are gathered closely may pose challenges to instance segmentation.

Based on the instance segmentation process, individual buildings can be obtained. Subsequently, the geo-referenced LiDAR point cloud is registered with the corresponding as-designed BIM model using the proposed matching strategy, with the LiDAR data serving as references. Once the LiDAR data and BIM models are matched, all as-designed BIM models will obtain geo-referenced information, resulting in the formation of urban GeoBIM models (Fig. 11).

TABLE II
POSITIONING ACCURACY OF BUILDING BIM MODELS. "N" IS THE NUMBER OF CORRESPONDING OBJECTS

Dataset	Building positioning errors (m)				
	N	Min	Max	Avg.	RMSE
1	3	0.15	0.21	0.18	0.18
2	10	0.09	0.18	0.13	0.14

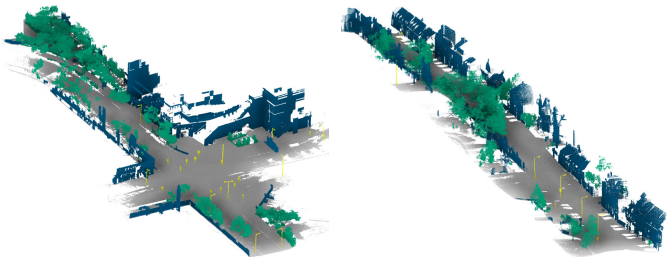


Fig. 12. Training samples labeling in (a) Dataset 1 and (b) Dataset 2.

C. Urban GeoBIM Construction

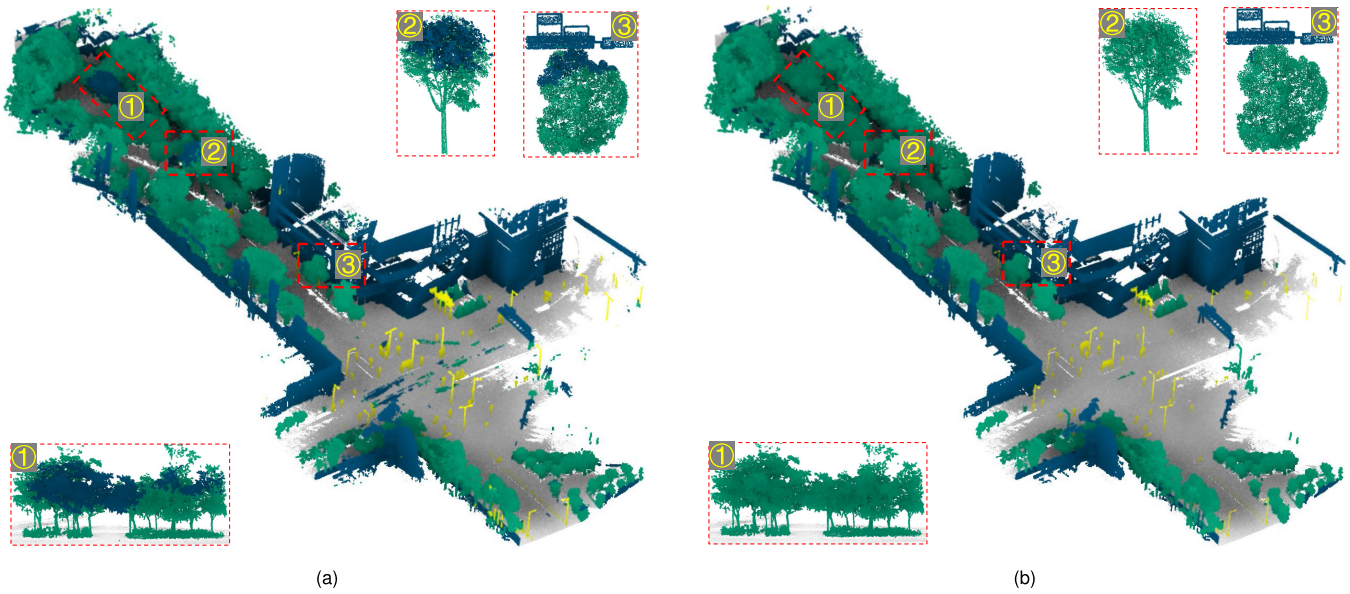
The construction of urban GeoBIM models has resulted in a more accurate representation of 3-D real cities in the form of geo-referenced BIM models. The spatial distribution of as-designed BIM models for all buildings is consistent with that of their corresponding LiDAR point clouds, indicating the correctness and effectiveness of the matching process. To quantitatively evaluate the construction accuracy, the geo-referenced positions of all BIM models are compared with those of the corresponding LiDAR data. The position distances between BIM models and LiDAR data are calculated and presented in Table II for further analysis and assessment of the constructed GeoBIM models. From the results, building positioning accuracies varied between 0.05 and 0.25 m, with buildings having complex structures generally exhibiting high positioning accuracy owing to rich structural information. Overall, the proposed method has the potential to construct accurate urban GeoBIM models.

V. DISCUSSION

A. Analysis of Training Sample Labeling

Training samples used in the supervised classification network are critical inputs that determine the effectiveness and accuracy of the semantic segmentation in this study. The results obtained so far have demonstrated the reliability of the training samples. As shown in Fig. 12, the training samples cover a wide range of urban scenes, and each category exhibited accurate labeling results. This not only provides reliable inputs for supervised semantic segmentation automatically but

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(a) (b)

Fig. 13. Segmentation optimization of Dataset 1. Red rectangle in (a) shows some vegetation points (blue points) that were misclassified as buildings. (b) Corrected results.

also takes into consideration the topological and geometric features at both global and local levels. This indicates the robustness and effectiveness of the proposed approach for semantic segmentation of LiDAR point clouds.

The labeling precision of training samples is analyzed based on the criteria outlined in Section III-E (Table III). It is found that nearly half of the LiDAR point cloud can be automatically labeled, providing support for the weakly supervised semantic segmentation network to obtain more reliable results. Furthermore, the labeling precision for each category is high, exceeding 80% in the two datasets. Ground and pole-like categories exhibit particularly high labeling precision, reaching around 95% owing to their unique geometric characteristics (elevation for ground and line features for pole-like objects). Buildings and vegetation categories can be generally distinguished based on their planar characteristics, with precisions of approximately 85%. However, some irregular objects adjacent to vegetation, such as pedestrians and fences, may be mislabeled as the vegetation category due to varying the orientation of local surface points. Similarly, some regular objects adjacent to buildings, such as cars and tree trunks, may also be mislabeled as buildings due to local surface consistency characteristics. Nevertheless, the percentage of mislabeled points is small overall, and a large percentage of correctly labeled object points can ensure semantic segmentation accuracy.

TABLE III
LABELING PRECISION (%) OF TRAINING SAMPLES

Dataset	Classes			
	Ground	Building	Vegetation	Pole-like
1	94.58	82.03	85.81	98.65
2	98.51	83.08	96.47	96.79

B. Analysis of Graph-Based Clustering

Moreover, the proposed graph-based clustering method can also be used to optimize semantic segmentation results, in addition to its application in instance segmentation. In general, the weak supervision framework may produce false segmentation points due to the complexity of city scenes. For instance, vegetation points misclassified as buildings constitute the majority of false segmentation, and some internal points of buildings may be segmented as vegetation due to discrete characteristics (refer to Fig. 13).

The results of the proposed graph-based clustering method show that misclassified points, which are generally in the minority and adjacent to correctly segmented points, can be corrected using the distance information between adjacent objects, as shown in Fig. 13(a). However, for points that are adjacent to multiple objects at the same time, it is challenging to accurately determine their final class based solely on distance information, as shown in Fig. 13(b). To address this issue, the proposed clustering method integrates roughness and anisotropy information to optimize the incorrectly segmented points located at object adjacency, leveraging the similarity of local structural properties on the same object.

Additionally, the method removes outliers that are characterized by discreteness and isolation, which effectively improves the visualization performance of semantic point clouds. As a result, the segmentation accuracy is improved compared to the semantic segmentation results, with a notable improvement of 3.3% in overall accuracy and 1.96% in average F1 score improved for Dataset 1, as shown in Table IV. This demonstrates the effectiveness of the graph-based clustering process in refining and improving the segmentation accuracy, particularly in cases where misclassified points are adjacent to multiple objects or characterized by discreteness and isolation.

C. Analysis of Instance Segmentation

The quantitative analysis of the extracted buildings and pole-like objects from Dataset 2 is further delineated in

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TABLE IV
CONFUSION MATRIX OF CLUSTERING RESULTS WITH PER-CLASS PRECISION, RECALL, AND F1 SCORE (%)

Dataset	Criteria (%)	Classes					Avg. F1 (%)	OA (%)
		Others	Ground	Building	Vegetation	Pole-like		
1	Precision	99.08	96.75	77.62	97.81	98.52	86.96	93.78
	Recall	42.00	99.94	96.06	99.43	88.07		
	F1 score	59.00	98.32	85.86	98.61	93.00		
2	Precision	87.61	99.08	82.06	89.13	98.71	77.81	91.66
	Recall	37.52	99.29	98.37	93.40	39.69		
	F1 score	52.54	99.19	89.48	91.22	56.61		

TABLE V
QUANTITATIVE ANALYSIS OF INSTANCE SEGMENTATION RESULTS OF DATASET 2. PLEASE NOTE THAT COMPARED METHODS WERE STUDIED IN OTHER DATASETS

Category	Method	Completeness (%)	Correctness (%)	Quality (%)
Building	Zhang et al.	100.00	100.00	100.00
	Ours	100.00	100.00	100.00
Pole	Tu et al.	91.23	90.43	83.20
	Ours	47.46	100.00	47.46

Table V. Three popular evaluation criteria are taken into account, namely completeness, correctness, and quality. The concepts of completeness and correctness are analogous to recall and precision, respectively, while quality is calculated using the formula TP/(TP+FP+FN). A building segmentation method [49] and a pole extraction method [50] are utilized as reference points, notwithstanding the variations in the study areas. An examination of Table V reveals that the proposed method enables precise segmentation of buildings. Given the limited number of buildings in this scene and the distinct boundary structures, each building can be fully identified. Concerning pole-like objects, despite the extracted poles being accurately correct, the completeness value is relatively low (47.46%). This can be attributed to the fact that poles near buildings or trees can be easily misclassified as part of these structures in the semantic segmentation process, thereby posing significant challenges for instance segmentation and optimization. Consequently, investigating the precise semantics and instance segmentation of pole-like objects will be our future research on urban GeoBIM construction.

D. Matching Performance Evaluation

The core of the proposed urban GeoBIM construction is to provide geo-referenced information for as-designed BIM models in a local coordinate system by aligning them with LiDAR point clouds. Therefore, the calculation of the transformation between BIM models and LiDAR point clouds becomes a key aspect of this work. The matching results demonstrate that BIM models can align well with their corresponding LiDAR point clouds, with the building facades in both datasets showing consistent alignment. This indicated that the matching results have the potential to provide accurate geo-referenced information for BIM models, ensuring precise positioning in urban scenes. To highlight the advancements in matching, we compared the results of coarse alignment and fine registration obtained in this study with those achieved using the 4PCS and ICP algorithms, respectively (Table VI).

TABLE VI
MATCHING COMPARISON OF BUILDINGS IN DATASET 2

Errors (m)	Coarse alignment		Fine registration	
	4PCS	Ours	ICP	Ours
Avg.	0.90	0.62	0.37	0.13
RMSE	1.01	0.69	0.42	0.14

The comparison revealed that the proposed method demonstrated superior matching accuracies in both the coarse alignment and fine registration processes. In contrast to the 4PCS algorithm, the proposed method considered the optimal corresponding four-point pair, leading to more accurate results in the coarse alignment of LiDAR and BIM point clouds. While the ICP algorithm can reduce errors in coarse alignment, its efficacy fell short of the fine registration approach proposed in this study. The proposed method significantly reduced the error obtained by the ICP algorithm from approximately 0.4 m to around 0.1 m. These final errors may have been introduced due to the quality of BIM models and LiDAR data. In this study, BIM models were manually designed with reference to LiDAR data, which can be prone to human error and differences in LiDAR point cloud precision, resulting in inconsistencies between BIM models and LiDAR data, and subsequently affecting accurate correspondences. Additionally, LiDAR point clouds of incomplete buildings and sparse BIM model point clouds may result in low overlap, leading to inaccurate matching between LiDAR point clouds and BIM models. Therefore, if complete and high-precision LiDAR point clouds and as-designed BIM models were available, the matching accuracy could potentially be improved.

In general, ground-based LiDAR systems are capable of acquiring detailed structural information of urban scenes. However, due to limitations such as scanning location, the field of view, and measurement range, it is challenging for a single ground-based LiDAR system to capture complete urban scenes, resulting in incomplete building point clouds. This incomplete data makes it difficult to construct detailed BIM models from LiDAR point clouds directly. To overcome this challenge, the most common approach currently used for data collection is to combine multiple platforms and utilize multisource LiDAR point clouds, such as fusing airborne and ground-based LiDAR point clouds. However, compared to acquiring ground-based LiDAR data, acquiring airborne LiDAR point clouds is relatively knotty in urban scenes due to air restriction and safety factors. In modern infrastructure projects, detailed and precise BIM models are often designed prior to construction, and these rich BIM models can compensate for the lack of complete LiDAR data. It is in this context that we conducted this study.

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VI. Conclusion

In this study, we propose a novel urban GeoBIM construction approach that leverages the complementary advantages of geo-referenced LiDAR point clouds and as-designed BIM models. The approach includes semantic segmentation, instance segmentation, and matching of LiDAR data and BIM models. In the semantic segmentation step, an automated training sample labeling method is presented for adaptively extracting ground, buildings, pole-like objects, and vegetation. Then, a graph-based clustering technique by combining topological and geometric features is used to optimize semantic segmentation results and identify instance buildings and pole-like objects. Furthermore, LiDAR point clouds and as-designed BIM models are matched through the coarse alignment and fine registration processes for generating urban GeoBIM models. Experimental results demonstrate that the proposed approach improves segmentation accuracies, particularly in Dataset 1 where the Avg. F1 score is improved by 1.96%, from 85% to 86.9%. Moreover, the average positioning error obtained by the matching method is 0.156 m for buildings, indicating the potential of achieving detailed and accurate building GeoBIM construction in urban scenarios.

The proposed method is verified in small streets with different characteristics to demonstrate its general applicability. Future research will focus on extending the approach to regional or urban scales to provide comprehensive support for urban management and decision-making.

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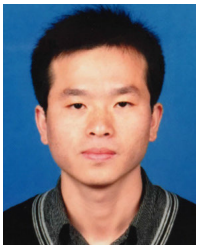
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Jie Shao received the Ph.D. degree from Beijing Normal University, Beijing, China, in 2020.

He is a Post-Doctoral Fellow with the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong. He is also an Assistant Research Fellow with The Hong Kong Polytechnic University Shenzhen Research Institute, Shenzhen, China.

His research interests include point cloud processing and deep learning.



Wei Yao (Senior Member, IEEE) received the B.S. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2003, and the Dipl.-Ing. (University) degree in geodesy and geoinformation and the Ph.D. degree in photogrammetry and remote sensing from the Technical University of Munich (TUM), Munich, Germany, in 2007 and 2010, respectively.

From 2007 to 2017, he was a Scientific Collaborator and a Guest Lecturer with the Institute of Photogrammetry and Cartography, TUM. From 2011 to 2017, he also worked as a Senior Scientist of research cluster of computer vision, remote sensing and navigation at Munich University of Applied Sciences, Munich. Since 2017, he has been a Professor at the Hong Kong Polytechnic University, Hong Kong. His research works were funded by research excellence program based on the Bavarian Elite Aid Act, German Federal Ministry for Environmental Protection (BMU), Sino-German Center for Research Promotion, National Natural Science Foundation of China (NSFC), Environmental Protection Department of Hong Kong SAR Government, and Research Grants Council (RGC) of Hong Kong. He has authored more than 130 academic articles in refereed international journals and conferences. His main research interests include multimodality remote sensing technology toward reconstruction and analysis of spatial–temporal behaviors of objects, image processing and analysis, machine learning, and related environmental and industrial applications.

Dr. Yao was a recipient of Best Presentation Award of the International Symposium on Mobile Mapping 2013 and multiple Best Paper Awards of IEEE/ISPRS events. He served as the Co-Chair of ISPRS WG III/6 for Remote Sensing Data Fusion as well as an editorial board/program committee members for multiple international/national journals and conferences in the field of LiDAR, photogrammetric image analysis, and remote sensing.



Puzuo Wang received the B.E. and M.E. degrees from Wuhan University, Wuhan, China, in 2017 and 2019, respectively. He is currently pursuing the Ph.D. degree with the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong.

His research interests include weakly supervised learning and domain adaptation for geospatial point cloud processing.



Zhiyi He received the B.E. and M.Sc. degrees from the College of Electrical and Informatic Engineering, Hunan University, Changsha, China, in 2018 and 2021, respectively.

He is currently working at the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong. His research interests include depth estimation and deep learning.



Lei Luo received the Ph.D. degree in cartography and geographic information system from the University of Chinese Academy of Sciences, Beijing, China, in 2016.

He is currently an Associate Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests focus on multiplatform earth observation technologies.

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